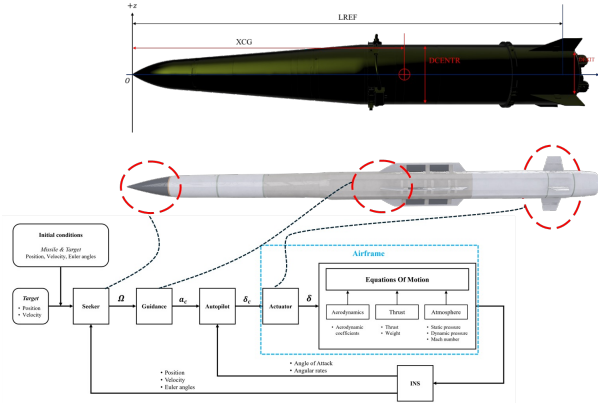
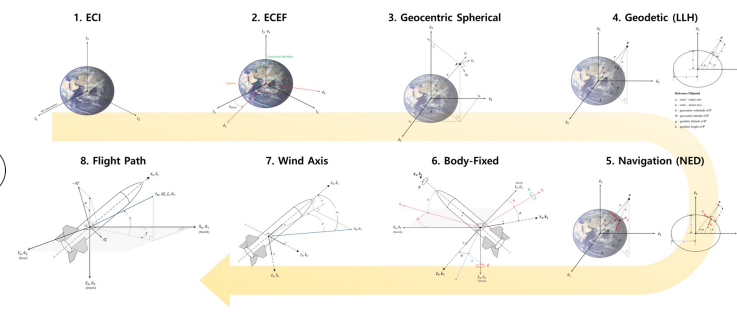
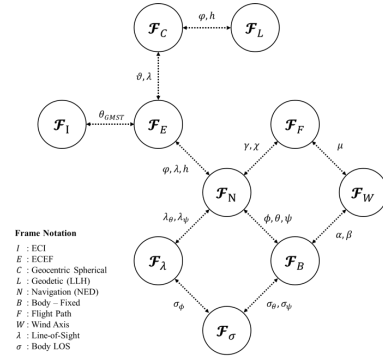


1. GNC Closed Loop

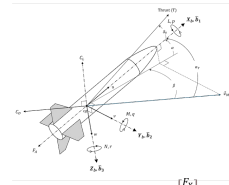


2. Coordinate Systems

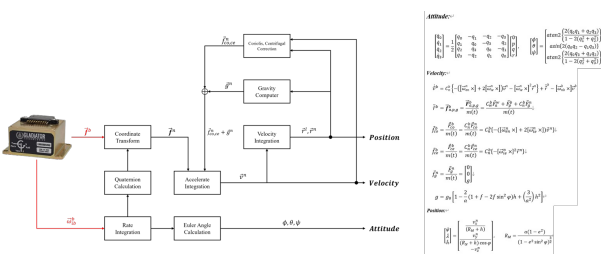


3. 6-DOF Missile Dynamics

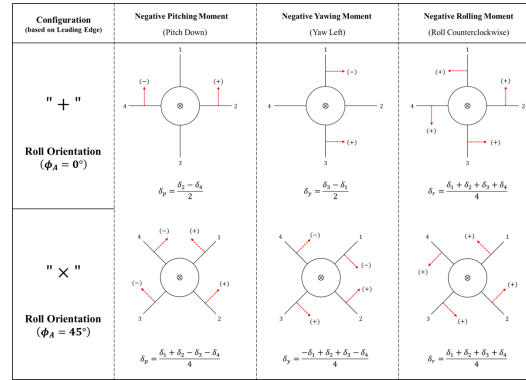
$$\begin{aligned} \frac{{}^n d\vec{L}}{dt} &= \frac{{}^b d\vec{L}}{dt} + {}^n \vec{\omega}^b \times \vec{L} = \vec{F}^b \\ \vec{F}^b &= \begin{bmatrix} u \\ v \\ w \end{bmatrix} = \begin{bmatrix} rv - qw + \frac{F_x}{m(t)} \\ pw - ru + \frac{F_y}{m(t)} \\ qu - pv + \frac{F_z}{m(t)} \end{bmatrix} \\ \vec{p} &= \frac{1}{I_{xx}} [L + (I_{yy} - I_{zz})qr] \\ \vec{q} &= \frac{1}{I_{yy}} [M + (I_{zz} - I_{xx})pr] \\ \vec{r} &= \frac{1}{I_{zz}} [N + (I_{xx} - I_{yy})pq] \end{aligned}$$



4. INS



5. Fin Configuration & Actuator



6. Ballistic Missile Simulation

